EB-context negative result and a strengthened AR baseline

Standalone addendum v1 to the frozen AMP H5 review package. The delivered v1 manuscript, supplement and source archive remain unchanged. This addendum records an already completed experiment; its preparation adds no training, tuning or sampling.

Decision and contribution boundary

The frozen terminal-conditioned partial-pooling candidate is **NOT_SUPPORTED**. A newly train-selected autoregressive (AR) baseline achieves lower development equal-terminal sequence NLL than the historical fixed AR and the candidate. This strengthens baseline assessment; it is not evidence of a new model contribution, general superiority or antimicrobial activity. The scientific submission decision remains **NO_GO**.

Frozen candidate and controls

The independent implementation uses a pooled, terminal-blind grouped-context tree with terminal-specific next-symbol distributions. Context grouping follows established PBCT ideas [1]; sharing a structure across processes is also prior art [2]. This is not an official PBCT reproduction. No downloaded author code was executed. For leaf e, terminal c, and one of 20 residues or EOS a, the candidate uses

$$p(a \mid e, c) = \frac{n_{e,c,a} + \lambda p_{\mathrm{pool},e}(a) + \alpha}{n_{e,c} + \lambda + 21\alpha}, \qquad p_{\mathrm{pool},e}(a) = \frac{\sum_c n_{e,c,a} + \alpha}{\sum_c n_{e,c} + 21\alpha}.$$

All counts and the pooled prior are fitted within each training fold. Pooled ignores terminal in emissions; separate uses independent terminal emissions on the same shared structure, not separately learned trees. The n-gram control uses terminal-specific longest-seen suffix backoff and add-α smoothing. Context lengths count preceding residues, with BOS padding at sequence starts.

Family	Frozen nine-choice grid
Pooled; separate	Each: depth $D \in \{1, 2, 3\}$; $\alpha \in \{0.1, 0.5, 1\}$.
Partial pooling	$D \in \{1, 2, 3\}$; $\lambda \in \{1, 10, 100\}$; fixed $\alpha = 0.5$.
Smoothed n-gram	Context length $D \in \{1, 2, 3\}$; $\alpha \in \{0.1, 0.5, 1\}$.
Train-CV AR	Learning rate $\{0.0003, 0.001, 0.003\}$; updates $\{250, 500, 1000\}$.

Tree learning fixes Dirichlet concentration 0.5, a one-nat penalty per additional child and minimum child count 8. Both AR models use the existing two-layer, 64-hidden-unit LSTM (62,501 parameters), with terminal input from BOS and every subsequent step. The fresh AR uses Adam, batch 32, gradient clipping 1, and training seed 222. It minimizes observed-token cross-entropy; selection uses full-sequence NLL.

Train-only selection and final evaluation order

Only train279 supplies the similarity graph: normalized global edit similarity at least 0.70, across both terminals, with transitive connected components. This gives 227 operational groups; each of three held-out folds contains 42 Free and 51 Amide sequences. These groups control near-duplicates, not proven biological homology. Every fit uses its fold's training rows; all out-of-fold scores are pooled before terminal averaging. Each family has nine selection opportunities; capacity and actual compute are not equal.

Selected configurations are partial ($D = 1, \alpha = 0.5, \lambda = 100$); pooled and separate ($D = 2, \alpha = 1$); n-gram ($D = 1, \alpha = 1$); fresh AR learning rate 0.0003 and 500 updates. Final models were frozen at 125.013 s; all generation finished at 125.108 s; the original dev CSV was first opened in this run at 125.109 s. Historical AR is an additional fixed, previously dev-exposed reference, not a fresh train-selected control.

Predictive and generation results

Common likelihood protocol and denominators

Every model scores all residues from BOS and the first EOS when length is below 48; no extra EOS is scored at the 48-residue cap. NLL is in nats per sequence. The primary summary averages sequences within each terminal, then gives Free and Amide equal weight. It is not an integrated PBCT marginal likelihood and does not omit the first D residues. All natural reference targets here are shorter than 48; the cap rule is validated by synthetic checks and retained capped generations.

Cohort and terminal	Sequences	Residues	EOS	Scored tokens
Train OOF: Free	126	2554	126	2680
Train OOF: Amide	153	2314	153	2467
Development: Free	18	330	18	348
Development: Amide	23	371	23	394

Full-sequence NLL; lower is better

Method	Grouped CV	Dev Free	Dev Amide	Dev equal
Partial pooling	51.239	53.002	45.327	49.164
Pooled	51.122	53.059	44.115	48.587
Separate emissions	51.299	52.939	44.897	48.918
Smoothed n-gram	51.570	53.461	45.487	49.474
Train-CV AR	49.870	51.752	44.334	48.043
Historical AR	—	62.898	53.035	57.967

Partial pooling fails the balanced-NLL comparison with pooled and fresh AR in grouped CV and with pooled, separate and fresh AR on dev. It slightly improves aggregate CV NLL over separate but worsens Amide CV NLL; therefore the frozen per-terminal gate still fails. The fresh AR has the lowest dev equal-terminal NLL among these controls; pooled has slightly lower Amide NLL. No statistical significance is asserted.

Raw generation: every entry is Free/Amide, each out of 32

Method	Empty	Length < 5	Cap 48	Valid 5–48
Partial pooling	0/0	3/11	2/0	29/21
Pooled	0/0	9/7	2/1	23/25
Separate emissions	0/0	5/13	3/0	27/19
Smoothed n-gram	0/1	5/12	2/0	27/20
Train-CV AR	2/0	2/1	1/0	30/31
Historical AR	0/0	0/0	0/0	32/32

Six methods each request 32 sequences per terminal: 384 attempts, no filtering, rejection or replenishment. Temperature is 1 and the maximum is 48. Seeds 422 and 423 define terminal streams for every method; Python and PyTorch random streams differ. Only terminal identity is supplied, with no reference sequence or oracle length. Empty outputs are included in the length-below-5 column. All 384 outputs have zero exact sequence copies of any training sequence; this does not rule out memorization generally. Fresh AR retains two empty Free outputs, so improved NLL is not uniform generation superiority.

Generation distances, budget and decision

Development distribution distances; Free/Amide, lower is better

Method	Linear MMD^2	Dipeptide JS	Length KS
Partial pooling	0.0064/0.0153	0.3348/0.3774	0.3576/0.5693
Pooled	0.0111/0.0125	0.3735/0.3779	0.3889/0.6318
Separate emissions	0.0091/0.0192	0.3663/0.4113	0.3819/0.6821
Smoothed n-gram	0.0090/0.0213	0.3666/0.4511	0.4444/0.6760
Train-CV AR	0.0053/0.0070	0.2894/0.3080	0.1889/0.3696
Historical AR	0.0058/0.0067	0.2945/0.2768	0.3229/0.1603

Distances use valid canonical lengths 1–48; request-level rates above retain empties. The frozen biased linear MMD^2 uses 421 length, residue and dipeptide features with equal block weights. JS is in bits; KS has no reported p-value. Partial is worse than fresh AR on all three listed distances in both terminals. The companion CSV retains both train and dev comparisons, AA JS, nearest-train identities and every method-terminal denominator. No metric is interpreted as biological efficacy.

Actual execution budget

There are 45 candidate configurations across five families, evaluated over three folds: **135 saved CV states**. These are not 135 independent training runs. They comprise 81 shared-tree emission states, 27 n-gram states and 27 AR checkpoints. The latter come from nine 1,000-update trajectories, scored at 250, 500 and 1,000 updates. Final fresh-AR fitting adds 500 updates: 9,500 new AR updates in total. Historical AR is loaded without refitting.

Six final models enter one fixed dev comparison stage and 384 generation requests. The original experiment used one CPU thread and serial model execution, with 1,000 s reserved for the runner and 200 s for independent recomputation. Actual runner time was **140.04 s**; all audit attempts took **37.03 s**; total was **177.07 s**, within the 1,200 s limit. One seed and zero natural-run retries were used. These are experiment-and-audit timings, not total human or document preparation time.

Frozen failure rules and observed decision

Partial had to improve equal-terminal NLL over every primary control in grouped CV and every dev control, including historical AR, while worsening neither terminal (tolerance 10^{-9}). Empty, short-below-5, cap-48 and exact-copy counts could not exceed pooled, separate, fresh AR or historical AR in either terminal. Invalid grouping, nonfinite values or budget failure would block execution. The observed predictive and short-output failures imply **NOT_SUPPORTED**; no grid enlargement or further tuning followed.

Independent checks and unchanged submission gates

Independent evidence checks

A non-author agent recomputed all 135 saved CV states and all six final-model dev likelihoods, rebuilt the selection and stop decisions, and verified every generation slot. It independently implemented EB probabilities and AR input/target construction. Legacy MMD, JS, length and nearest-neighbor formulas were rerun as shared-code consistency checks, not independently reimplemented. An auditor-only AR vocabulary error and JSON histogram-key comparison error were corrected; both failed receipts remain preserved. Natural code and results were not changed.

The original dev CSV was opened once after selection and model freezing. The later independent forward recomputation used the saved dev snapshot with no reselection. For this addendum, a separate non-author documentary review checks the existing records without any new model forward pass. Agent-based inconsistency checking does not constitute human peer review.

Limits and submission decision

The dev41 set was already reused during earlier H5 development. A final-only check within this experiment does not restore an untouched holdout or remove adaptive research-history bias. Similarity grouping is operational, model capacity and compute differ, and only one training seed is available. No test37 file was opened. This small frozen experiment supports a baseline correction and a negative candidate finding, not a new algorithm, learning-curve benefit, antimicrobial effect or submission readiness. Scientific **NO_GO** remains.

Bioinformatics policy block remains. The ISCB policy applicable to Bioinformatics prohibits LLM drafting of paper sections [3]. The existing review manuscript and this addendum involve substantive AI drafting. Disclosure, an AI rewrite or a private upload cannot cure that restriction. Current use remains **NOT_COMPLIANT_CURRENT_USE**. Human verification, author approval and compliant manuscript preparation remain unresolved.

Use of generative AI tools

Generative AI tools assisted with candidate implementation, experiment orchestration, evidence analysis, substantive drafting and LaTeX preparation. Separate agents performed implementation checks and record reconciliation. Human authors' verification and final approval have not been established. Authors retain responsibility for any future manuscript and its claims. This disclosure records actual use and grants no journal-policy clearance.

Sources and traceability

[1] Ghani, Heard and Sanna Passino. Approximate learning of parsimonious Bayesian context trees. Statistics and Computing 36, 106 (2026). Matched precedent for context grouping and reporting predictive comparisons.

[2] Approximate group context tree, arXiv:1107.0312. Matched methodological precedent for shared structure; no claim of independent novelty.

[3] ISCB Acceptable Use of Large Language Models Policy, updated 3 April 2025; accessed 12 October 2026.

The accompanying private evidence ZIP contains the frozen protocol, selected configs, summaries, request ledger, independent checks and hash bindings. Full model states remain in the local experiment archive.